

Update to the hake Reference Case Operating Model with the 2022 commercial and 2023 survey data

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Summary

Results for the 2023 RC update (which includes commercial CPUE and catch data up to 2022 and survey abundance estimates up to 2023) are compared to those for the 2022 RC. The 2022 and 2023 survey estimates and associated standard errors have been scaled by a factor of 1.25 to account for the fact that these surveys were conducted by industry vessel FV Compass Challenger, which has a different catchability coefficient to the *Africana*. The results for the 2023 RC are consistent with those from before the update. The 2023 estimates suggest that the *M. paradoxus* stock continues the slight downward trend which it has shown since 2020. This downward trajectory could still be considered reasonably compatible with abundance fluctuations evident for preceding years, but concern could be raised if indices continue to decline in the next few years. The *M. capensis* resource is estimated to be continuing on an upward trajectory.

Keywords: South African hake, Reference Case update, 2023 data

Introduction

Results are provided for two iterations of the Reference Case (RC) Operating Model (OM).

Run1: The 2022 RC. Note that this run is a slight update of the 2022 RC previously presented to the DWG (Run 3 of Ross-Gillespie, 2022), which did not take into account of the fact that the 2022 survey was conducted by an industry vessel and that the 2022 autumn survey estimates consequently should have been scaled by a factor of 1.25 (see Ross-Gillespie and Butterworth, 2018 and Winker, 2018). This has been corrected for Run 1 reported here. As the difference in results is negligible, results for the corrected run only are shown here.

Run2: An update of (1), with the 2022 commercial catch and CPUE data and the 2023 survey abundance estimates included in the model fit. The 2022 and 2023 survey estimates and associated standard errors have been scaled by a factor of 1.25 to account for the different catchability coefficient for the industry vessel FV Compass Challenger compared to the *Africana*.

Error! Reference source not found. lists the updated commercial catches for all the fleets from 1978 to 2022 used for Run2. Table 2 lists the updated GLM CPUE data used for Run2 and Table 3 lists the survey abundance estimates including the most recent 2023 survey estimates. The survey estimates reported in this Table have been scaled by the factor of 1.25.

Note that there was a glitch in the process of generating the 2022 GLM series used as input to the 2022 RC (Run1), resulting in the slightly higher values in that series than in previous years and the following year. It was not deemed necessary to re-do 2022 RC run with the corrected series for the results presented here, as it would likely have a minimal impact. Importantly, it should be noted that this glitch would not have impacted the 2023

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TAC recommendation as the OMP-formula standardises these data relative to the 2011-2020 values, so that the difference in magnitude had a minimal impact on the OMP formula output and no impact on the final TAC advice.

Results

Table 4 lists key assessment outputs for the 2022 and 2023 RCs. Table 5a and b list the negative log-likelihood components for the two models, in (a) absolute terms and (b) relative to Run1.

Figure 1 - Figure 6 show various assessment outputs for the two model fits.

Discussion

The results for the RC run with the updated data are consistent with those from before the update. The updated B^{sp}/B_{MSY} estimate for 2023 is slightly lower for *M. paradoxus* than that for 2022 (1.59 compared to 1.65, see Table 4) and slightly higher for *M. capensis* (2.82 compared to 2.68). The 2023 estimates suggest that the *M. paradoxus* stock continues the downward trend which it has shown since 2020. This downward trajectory could still be considered reasonably compatible with fluctuations evident for preceding years, but concern could be raised if indices continue to decline in the next few years. The *M. capensis* resource is estimated to be continuing on an upward trajectory.

References

- Ross-Gillespie, A. 2022. Update to the hake Reference Case Operating Model with corrected longline data, and 2021 commercial and 2022 survey data. DEFF Fisheries document: FISHERIES/2022/OCT/SWG-DEM/35rev: 15pp.
- Ross-Gillespie, A. and Butterworth, D.S. 2018. First set of robustness tests conducted for OMP2018. International Stock Assessment Workshop document MARAM/IWS/2018/Hake/P6arev.
- Fairweather, T.P. 2023. Standardised abundance estimates for key commercial species 2016-2023. DFFE Fisheries document: FISHERIES/2023/SEP/SWG-DEM/06: 2pp.
- Winker, H. 2018. A state-space approach to approximate changes trawl survey catchability for *Merluccius paradoxus* and *M. capensis*. DFFE document FISHERIES/2018/SEP/SWG-DEM/45. Also International Stock Assessment Workshop document MARAM/IWS/2018/Hake/BG5.

Table 1: Species-disaggregated catches (in thousand tons) by fleet of South African hake from the south and west coasts for the period 1978-present. The 2022 catches were provided by D. Durholtz (*pers. comm.*). Grey highlighting has been used to mark updated and new catch data. Note that after the presentation of this document to the DWG, it was realised that the full TAC had not been caught in 2022. Specifically, there were under-catches of 7518 tons for offshore and 3547 tons for inshore. Some eelbooks had been omitted which totalled 3351 tons of offshore catch. The remaining difference between the landed catch and TAC (4168 tons) represents less than 4% of the offshore allocation and the inshore sector routinely under-catches its quota. The analyses presented here have not been updated at this stage to include the catches omitted, but their impact will be negligible.

	<i>M. paradoxus</i>				<i>M. capensis</i>						Total
	Offshore		Longline		Offshore		Inshore	Longline		Handline	
	WC	SC	WC	SC	WC	SC	SC	WC	SC	SC	
1978	101.042	3.220	-	-	26.470	4.365	4.931	-	-	-	140.028
1979	94.331	1.924	-	-	39.192	4.995	6.093	-	-	-	146.535
1980	99.654	2.206	-	-	33.873	4.254	9.121	-	-	-	149.108
1981	88.883	0.910	-	-	32.048	4.575	9.400	-	-	-	135.816
1982	83.618	3.353	-	-	29.732	8.005	8.089	-	-	-	132.797
1983	71.238	4.723	0.126	-	23.195	7.792	7.672	0.104	-	-	114.850
1984	82.358	3.796	0.200	0.005	28.897	7.139	9.035	0.166	0.011	-	131.607
1985	94.428	8.059	0.638	0.091	30.642	11.957	9.203	0.529	0.201	0.065	155.813
1986	103.756	8.580	0.753	0.094	30.049	7.385	8.724	0.625	0.208	0.084	160.258
1987	93.517	7.460	1.952	0.110	24.008	8.225	8.607	1.619	0.243	0.096	145.837
1988	79.913	5.876	2.833	0.103	26.669	8.640	8.417	2.350	0.228	0.071	135.100
1989	82.230	6.182	0.158	0.010	25.029	12.730	10.038	0.132	0.022	0.137	136.668
1990	81.996	9.341	0.211	-	21.640	13.451	10.012	0.175	-	0.348	137.174
1991	87.093	12.448	-	0.932	19.357	9.626	8.206	-	2.068	1.270	141.000
1992	84.768	17.297	-	0.466	18.519	9.165	9.252	-	1.034	1.099	141.600
1993	102.125	9.880	-	-	15.940	4.380	8.870	-	-	0.278	141.473
1994	103.541	6.726	0.882	0.194	20.327	4.326	9.569	0.732	0.432	0.449	147.178
1995	100.268	4.004	0.523	0.202	20.629	3.146	10.630	0.434	0.448	0.756	141.040
1996	107.381	8.966	1.308	0.568	21.794	4.323	11.062	1.086	1.260	1.515	159.263
1997	100.654	10.509	1.410	0.582	16.500	5.327	8.834	1.170	1.290	1.404	147.680
1998	111.154	9.742	0.505	0.457	16.499	4.411	8.283	0.419	1.014	1.738	154.222
1999	88.581	11.420	1.532	1.288	15.179	3.926	8.595	1.272	2.856	2.749	137.398
2000	96.587	7.700	2.706	3.105	21.114	5.830	10.906	2.000	1.977	5.500	157.425
2001	101.247	7.850	1.417	0.084	16.349	8.306	11.836	2.394	1.527	7.300	158.310
2002	91.207	12.443	4.469	1.585	13.724	6.141	9.581	2.391	2.546	3.500	147.587
2003	93.711	17.397	3.305	1.252	11.665	7.636	9.883	2.526	3.078	3.000	153.453
2004	85.722	26.065	2.855	1.196	12.510	8.704	10.004	2.297	2.731	1.600	153.684
2005	85.869	21.778	3.091	0.472	9.398	7.468	7.881	2.773	3.270	0.700	142.700
2006	81.513	18.050	3.241	0.485	11.984	6.578	5.524	2.520	3.227	0.400	133.522
2007	92.724	13.488	2.512	3.021	16.145	3.757	6.350	2.522	2.522	0.400	143.441
2008	85.538	13.191	2.255	0.809	13.838	4.316	5.496	1.937	1.893	0.231	129.504
2009	68.202	10.895	2.410	1.069	12.296	4.806	5.639	2.828	2.520	0.265	110.930
2010	69.709	15.457	2.394	1.527	10.186	4.055	5.472	3.086	3.024	0.275	115.185
2011	76.576	17.904	2.522	0.140	15.673	4.086	6.013	3.521	3.047	0.186	129.668
2012	81.411	16.542	4.358	0.306	12.928	4.584	3.223	2.570	1.737	0.008	127.667
2013	74.341	28.859	6.056	0.060	8.761	4.475	2.920	2.606	1.308	0.000	129.386
2014	73.252	41.156	6.879	0.008	9.671	6.286	2.965	2.123	0.315	0.001	142.656
2015	77.521	31.745	5.205	0.023	12.727	4.035	3.077	3.025	0.069	0.001	137.428
2016	93.173	18.968	3.697	0.001	14.744	2.810	3.973	5.745	0.003	0.001	143.115
2017	72.326	30.961	5.300	0.025	15.273	4.466	2.812	2.813	0.126	0.004	134.106
2018	64.252	29.218	5.217	0.09	12.689	12.863	3.983	2.646	0.487	0.024	131.469
2019	72.176	20.880	5.328	0.034	14.508	8.892	4.148	3.623	0.299	0.009	129.896
2020	94.453	11.731	5.847	0.047	16.513	4.081	3.872	2.348	0.321	0.004	139.218
2021	101.262	14.764	5.892	0.018	17.060	3.484	4.517	2.932	0.194	0.010	150.132
2022	71.645	11.847	5.541	0.019	16.074	3.364	4.584	2.368	0.077	0.001	115.520

Table 2: GLM standardized CPUE data for *M. paradoxus* and *M. capensis* provided by J. Glazer (*pers. comm.*).

Year	GLM CPUE (kg min ⁻¹)			
	<i>M. paradoxus</i>		<i>M. capensis</i>	
	West Coast	South Coast	West Coast	South Coast
1978	8.77	1.39	2.35	3.98
1979	8.82	1.13	3.25	5.06
1980	8.26	2.12	2.92	5.43
1981	8.10	1.33	3.04	4.69
1982	7.90	2.02	2.63	4.94
1983	8.76	2.03	3.28	5.81
1984	8.97	2.26	3.34	6.74
1985	10.54	3.05	3.75	8.49
1986	9.12	3.08	3.10	6.98
1987	7.62	2.83	2.62	6.41
1988	7.40	2.27	2.39	6.53
1989	8.02	2.37	2.56	7.28
1990	8.49	3.13	2.29	8.77
1991	9.50	3.47	2.78	7.84
1992	8.46	3.86	3.19	7.36
1993	8.54	2.97	3.07	5.43
1994	9.13	2.85	3.61	6.50
1995	8.11	2.14	3.47	6.00
1996	8.95	3.18	3.98	6.00
1997	7.92	3.43	3.37	5.01
1998	9.01	3.23	3.83	4.94
1999	7.39	3.83	3.42	5.35
2000	6.68	3.38	3.22	5.96
2001	5.34	2.88	2.28	4.49
2002	5.22	2.59	2.18	4.82
2003	6.01	3.02	1.95	5.16
2004	4.92	2.52	1.81	4.29
2005	4.79	2.16	1.38	3.69
2006	5.08	2.25	1.57	2.91
2007	6.19	2.36	1.65	2.85
2008	6.75	2.43	2.04	3.75
2009	6.92	2.89	2.94	6.43
2010	7.75	3.19	2.65	5.18
2011	7.61	3.71	3.14	5.91
2012	6.63	3.29	2.66	3.87
2013	6.62	3.67	2.72	4.81
2014	6.69	3.45	2.24	2.82
2015	8.68	3.27	2.77	2.87
2016	8.77	3.38	2.76	3.80
2017	7.91	3.73	2.84	5.16
2018	6.94	4.19	2.93	9.31
2019	6.31	3.19	2.86	6.97
2020	7.37	3.16	3.28	4.91
2021	7.88	3.90	3.83	5.67
2022	5.94	2.79	2.89	3.88

Table 3a: Survey abundance estimates and associated standard errors in thousand tons for *M. paradoxus* for the depth range 0-500m for the South Coast and for the West Coast (Fairweather, 2023). As the 2022 and 2023 surveys were conducted by the industry vessel FV Compass Challenger, the estimates shown here (marked with an asterisk) have already been multiplied by a factor of 1.25 to account for different catchability coefficients.

Year	West coast				South coast			
	Summer		Winter		Spring (Sept)		Autumn (Apr/May)	
	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)
1985	168.989	(37.765)	290.281	(63.295)	-	-	-	-
1986	202.334	(37.745)	147.378	(21.667)	11.280	(3.111)	-	-
1987	284.434	(54.165)	180.158	(39.047)	16.381	(3.033)	-	-
1988	138.534	(20.303)	252.121	(71.246)	-	-	28.293	(8.673)
1989	-	-	434.092	(142.716)	-	-	-	-
1990	307.615	(87.841)	205.704	(43.607)	-	-	-	-
1991	331.177	(81.633)	-	-	-	-	27.570	(8.153)
1992	225.755	(33.711)	-	-	-	-	25.036	(6.650)
1993	340.079	(51.427)	-	-	-	-	162.375	(81.691)
1994	333.499	(56.259)	-	-	-	-	108.179	(38.369)
1995	317.104	(76.709)	-	-	-	-	70.890	(39.330)
1996	474.270	(92.744)	-	-	-	-	68.859	(19.929)
1997	543.615	(96.043)	-	-	-	-	121.707	(51.507)
1998	-	-	-	-	-	-	-	-
1999	542.830	(110.541)	-	-	-	-	263.256	(59.439)
2000	-	-	-	-	-	-	-	-
2001	-	-	-	-	16.668	(7.159)	-	-
2002	251.820	(32.690)	-	-	-	-	-	-
2003	386.321	(63.565)	-	-	98.434	(42.249)	185.345	(82.188)
2004	271.540	(55.710)	-	-	70.001	(22.156)	39.822	(22.153)
2005	296.065	(42.409)	-	-	-	-	26.691	(6.017)
2006	316.247	(57.332)	-	-	68.507	(18.283)	34.868	(5.843)
2007	407.377	(77.222)	-	-	66.267	(21.966)	102.195	(53.688)
2008	238.143	(37.018)	-	-	25.661	(8.324)	33.034	(9.340)
2009	310.760	(27.768)	-	-	-	-	45.030	(15.551)
2010	576.848	(88.202)	-	-	-	-	46.938	(12.160)
2011	380.185	(128.013)	-	-	-	-	21.054	(6.531)
2012	405.865	(59.099)	-	-	-	-	-	-
2013	136.260	(25.116)	-	-	-	-	-	-
2014	269.482	(37.492)	-	-	-	-	62.925	(24.802)
2015	207.583	(24.057)	-	-	-	-	111.411	(51.852)
2016	312.876	(33.250)	-	-	-	-	94.177	(51.731)
2017	319.024	(58.766)	-	-	-	-	-	-
2018	-	-	-	-	-	-	-	-
2019	243.560	(51.558)	-	-	-	-	-	-
2020	243.090	(43.989)	-	-	-	-	33.176	(15.444)
2021	-	-	-	-	-	-	50.060	(20.430)
2022	318.394*	(85.361)*	-	-	-	-	-	-
2023	133.879*	(17.571)*	-	-	-	-	19.826*	(4.678)*

Table 3b: Survey abundance estimates and associated standard errors in thousand tons for *M. capensis* for the depth range 0-500m for the South Coast and for the West Coast (Fairweather, 2023). As the 2022 and 2023 surveys were conducted by the industry vessel FV Compass Challenger, the estimates here (marked with an asterisk) have already been multiplied by a factor of 1.25 to account for different catchability coefficients.

Year	West coast				South coast			
	Summer		Winter		Spring (Sept)		Autumn (Apr/May)	
	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)	Biomass	(s.e.)
1985	102.929	(18.888)	159.198	(18.982)	-	-	-	-
1986	113.154	(23.474)	115.218	(19.733)	96.768	(10.737)	-	-
1987	75.438	(9.709)	83.050	(10.306)	137.008	(13.057)	-	-
1988	66.365	(9.930)	48.046	(9.574)	-	-	154.548	(23.984)
1989	-	-	294.740	(67.495)	-	-	-	-
1990	400.142	(97.102)	156.337	(22.507)	-	-	-	-
1991	67.565	(9.656)	-	-	-	-	276.607	(25.274)
1992	95.401	(11.892)	-	-	-	-	124.495	(13.600)
1993	93.613	(14.390)	-	-	-	-	144.551	(12.379)
1994	124.497	(37.845)	-	-	-	-	153.790	(20.310)
1995	193.292	(24.270)	-	-	-	-	222.464	(31.245)
1996	87.969	(9.866)	-	-	-	-	222.176	(23.144)
1997	252.606	(42.721)	-	-	-	-	163.163	(17.274)
1998	-	-	-	-	-	-	-	-
1999	188.624	(31.362)	-	-	-	-	171.946	(13.330)
2000	-	-	-	-	-	-	-	-
2001	-	-	-	-	117.590	(20.093)	-	-
2002	105.093	(16.130)	-	-	-	-	-	-
2003	73.020	(12.518)	-	-	73.604	(9.142)	117.538	(17.192)
2004	194.294	(30.714)	-	-	96.933	(13.936)	92.796	(11.318)
2005	63.363	(11.498)	-	-	-	-	68.672	(5.302)
2006	73.655	(17.255)	-	-	92.831	(8.998)	116.298	(11.931)
2007	73.230	(9.306)	-	-	67.937	(6.553)	65.935	(5.303)
2008	52.577	(7.069)	-	-	87.836	(9.723)	102.169	(9.681)
2009	140.437	(26.486)	-	-	-	-	111.191	(10.832)
2010	162.402	(34.891)	-	-	-	-	170.261	(33.235)
2011	89.095	(23.574)	-	-	-	-	105.424	(10.688)
2012	84.746	(8.331)	-	-	-	-	-	-
2013	30.383	(4.575)	-	-	-	-	-	-
2014	219.756	(60.342)	-	-	-	-	63.389	(6.415)
2015	65.086	(9.178)	-	-	-	-	76.059	(6.873)
2016	115.058	(30.400)	-	-	-	-	83.197	(6.600)
2017	69.289	(14.486)	-	-	-	-	-	-
2018	-	-	-	-	-	-	-	-
2019	62.560	(7.697)	-	-	-	-	-	-
2020	109.983	(11.836)	-	-	-	-	132.099	(14.486)
2021	-	-	-	-	-	-	89.119	(10.798)
2022	175.716*	(37.074)*	-	-	-	-	-	-
2023	238.281*	(70.323)*	-	-	-	-	109.256*	(16.018)*

Table 4: Key assessment outputs for the two OMs. Biomass units are thousand mt.

<i>M. paradoxus</i>	Run1	Run2	<i>M. capensis</i>	Run1	Run2
K^{sp}	458	470	K^{sp}	239	237
B_{MSY}	74	72	B_{MSY}	70	70
B_{MSY} / K^{sp}	0.16	0.15	B_{MSY} / K^{sp}	0.29	0.30
MSY	136	135	MSY	83	83
B_{2022}^{sp}	111	119	B_{2022}^{sp}	196	188
B_{2022}^{sp} / K^{sp}	0.24	0.25	B_{2022}^{sp} / K^{sp}	0.82	0.79
B_{2022}^{sp} / B_{MSY}	1.50	1.65	B_{2022}^{sp} / B_{MSY}	2.80	2.68
B_{2023}^{sp}	-	115	B_{2023}^{sp}	-	197
B_{2023}^{sp} / K^{sp}	-	0.24	B_{2023}^{sp} / K^{sp}	-	0.83
B_{2023}^{sp} / B_{MSY}	-	1.59	B_{2023}^{sp} / B_{MSY}	-	2.82

Table 5a: Negative log-likelihood values for fits to the two OMs. The values for Run2 in grey text indicate that those negative log-likelihood components are not comparable to those from the 2022 RC on account of data updates.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
Run1: 2022 RC	-242.68	-39.06	-41.80	-1573.56	-1652.95	9.69	128.81	0.02	-3411.53
Run2: 2023 data update	-239.51	-39.33	-40.92	-1572.62	-1652.19	9.82	128.70	0.02	-3406.04

Table 5b: The first row of this Table lists the negative log-likelihood for the 2022 RC, while the second row shows the difference between the 2022 RC and the 2023 update. A positive value indicates a worse negative log-likelihood for the update, and those cells have also been shaded in grey. Grey text and dashed lines have been used to indicate the likelihood components for Run2 which are not comparable with those from Run1, on account of data updates.

Run	GLM CPUE	ICSEAF CPUE	Survey abun.	Comm. CAL	Survey CAL	Recruit. resid.	ALKs	Penalties	Total (w/o pen.)
Run1: 2022 RC	-242.64	-39.08	-42.28	-1573.51	-1652.96	9.60	128.83	0.02	-3412.02
Run2: 2023 data update	3.12	-0.25	1.36	0.88	0.77	0.22	-0.13	0.00	5.98

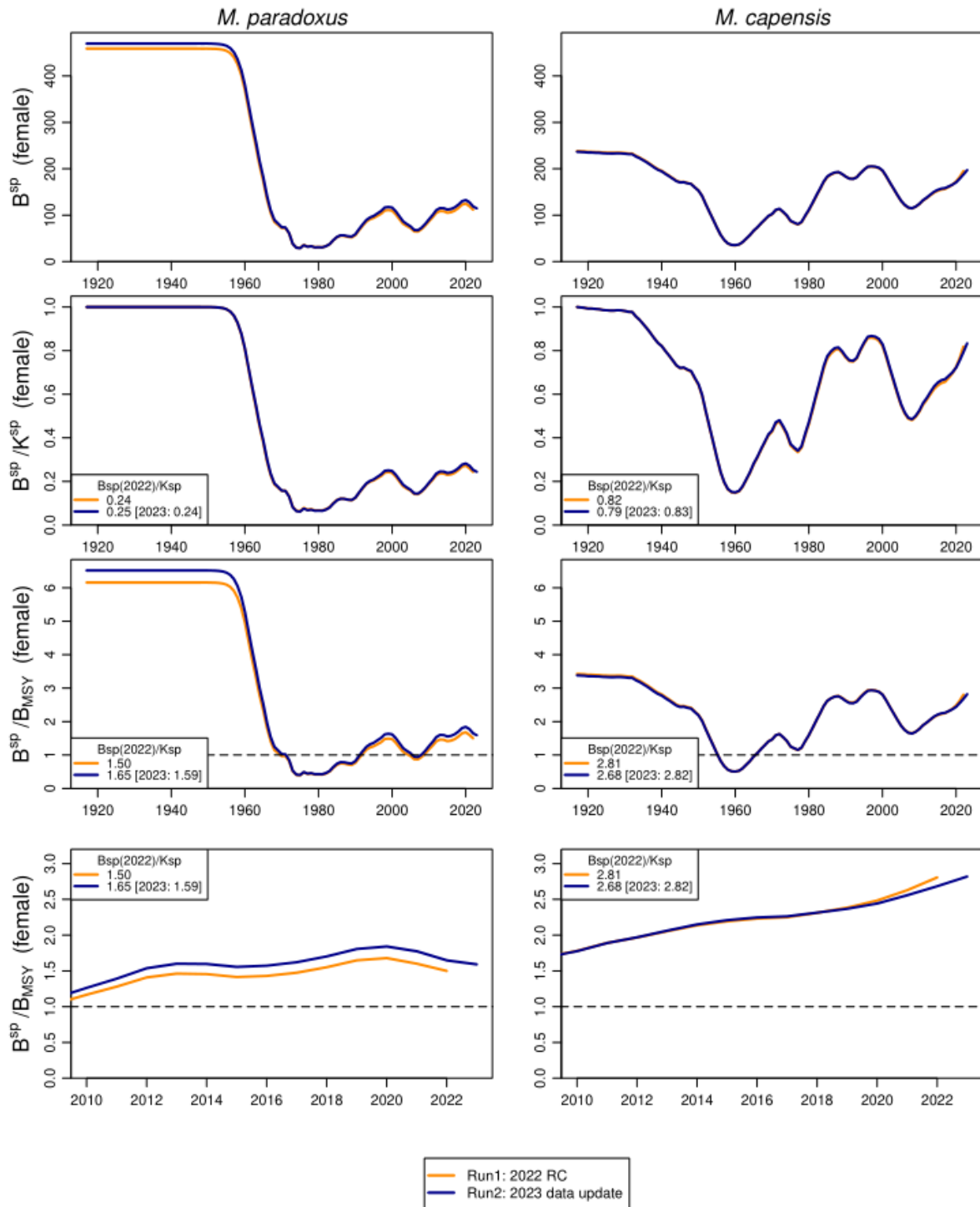


Figure 1: Female spawning biomass is shown for each species in absolute terms (top row), relative to pristine spawning biomass (second row), relative to B_{MSY} (third row) and relative to B_{MSY} but for the 2010-2023 time period only (fourth row). The depletion levels have been listed in the figure legends for the year 2022, with the 2023 values for Run2 included in square brackets.

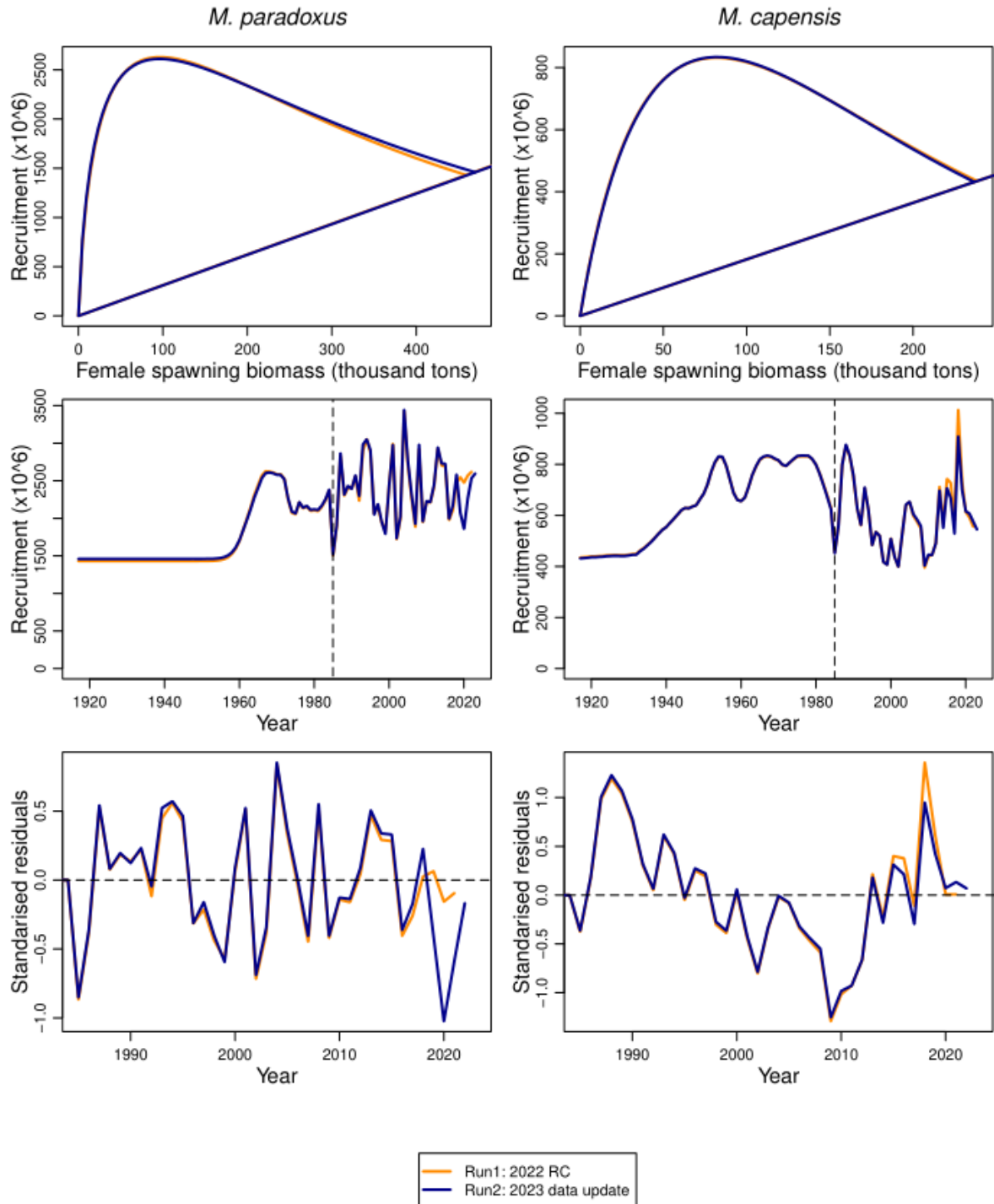


Figure 2: The top row shows the estimated recruitment-spawning biomass relationship for the two assessment models, while the second row shows the estimated recruitment time-series values and the bottom row the estimated standardized recruitment residuals.

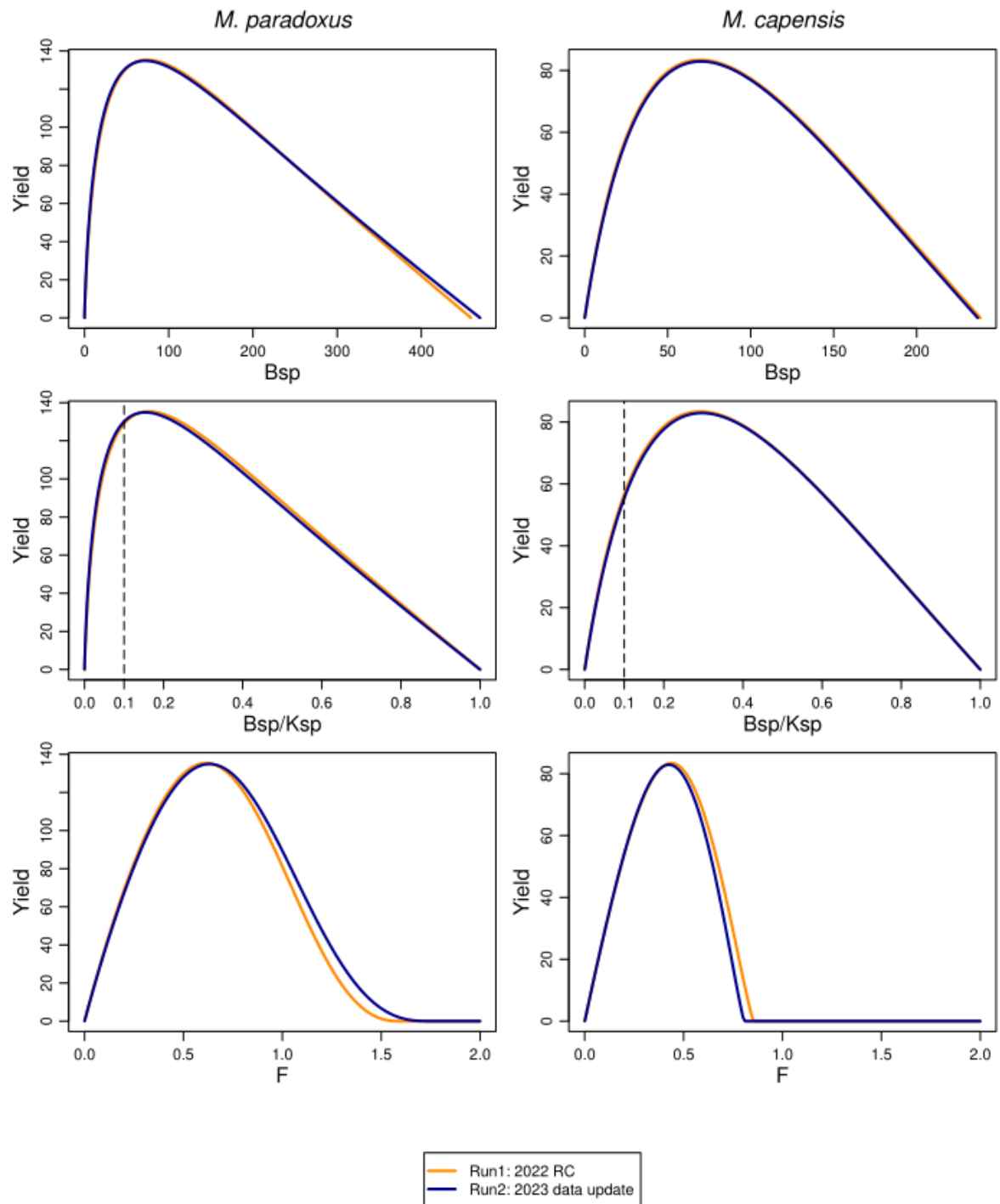


Figure 3: The estimated sustainable yield curve (yield in thousand tons) plotted against female spawning biomass (top panel), female spawning biomass relative to pristine levels (middle panel) and fishing mortality rate F (bottom panel).

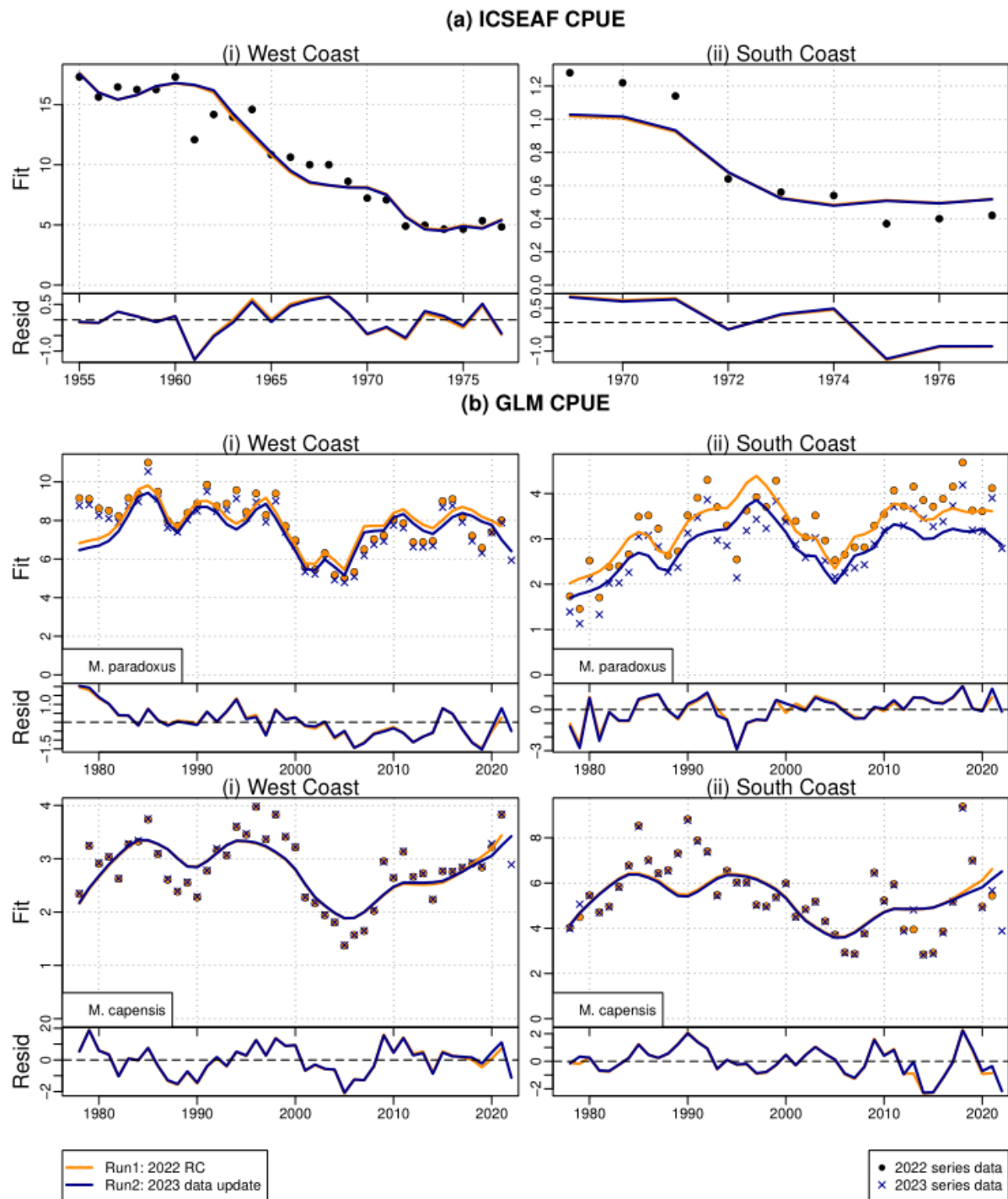


Figure 4: Fits to (A) the historical ICSEAF CPUE data and (B) the commercial GLM-standardized CPUE data are shown. Standardised residuals are shown underneath each plot. The 2022 GLM series are shown by yellow dots and the updated 2023 GLM CPUE series by the blue crosses. Note that there was a glitch in the process of generating the 2022 GLM series resulting in the slightly higher values in that series than in previous years and the following year. The corrected series (not shown here) is very close to the 2023 series. However, it was not deemed necessary to re-do the analyses with the corrected series. Note that this glitch would not have impacted the 2023 TAC recommendation as the OMP-formula standardises these data relative to the 2011-2020 values, so that the difference in magnitude had a minimal impact on the OMP formula output and no impact on the final TAC advice.

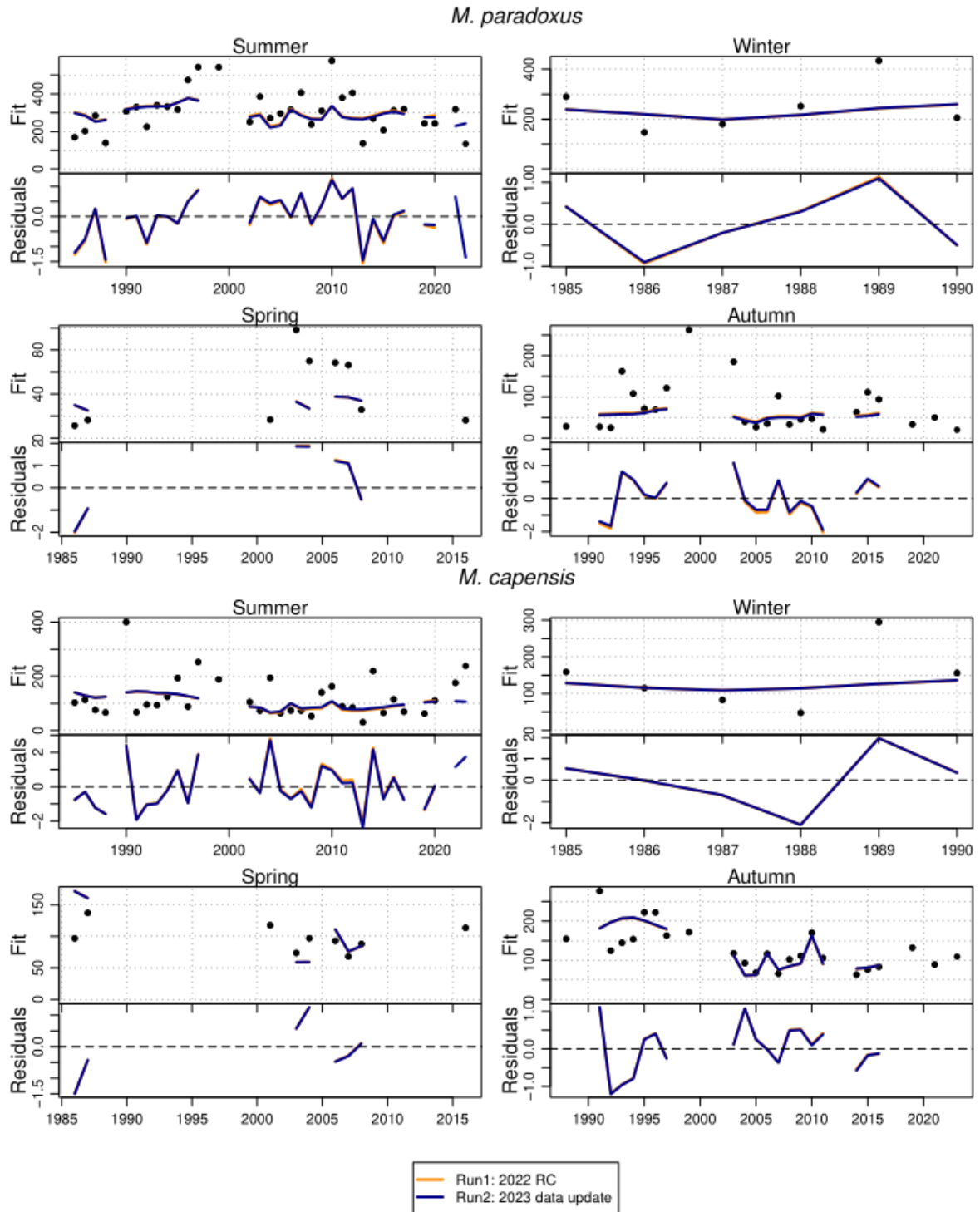


Figure 5: Fits to the survey relative abundance series are shown for the two OM. Standardised residuals are shown underneath each plot. Note that at this level of resolution, the two fits can scarcely be distinguished.

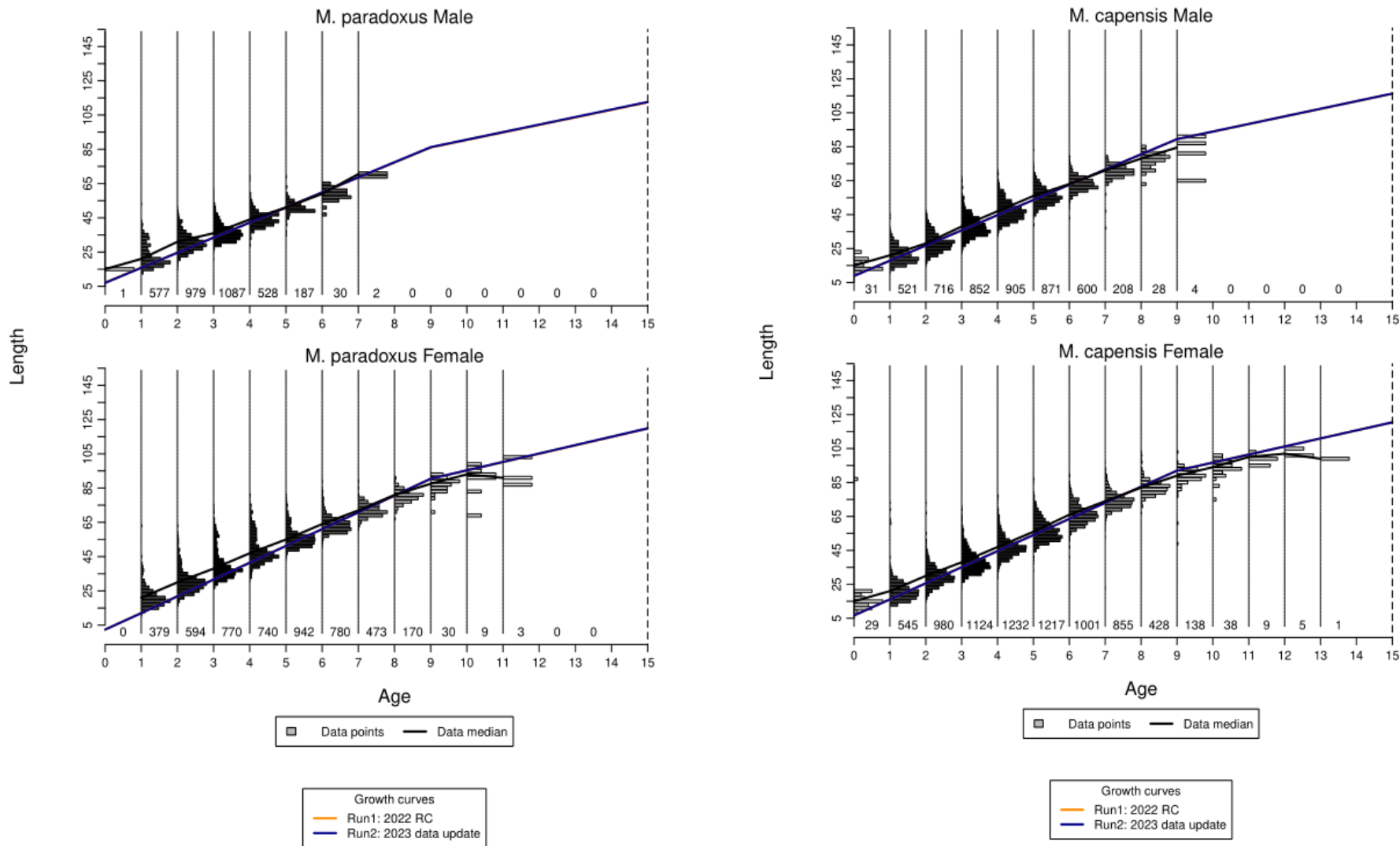


Figure 6: The estimated growth curves are superimposed on the length-at-age distributions derived from the age-length-key data. The numbers printed at the bottom of each age class indicate the number of data points available, and the shading of the bars is relative to these numbers, with darker shades indicating more data points.